

- 2 The cubic equation $2x^3 - 4x^2 + 3 = 0$ has roots α, β, γ . Let $S_n = \alpha^n + \beta^n + \gamma^n$.

(a) State the value of S_1 and find the value of S_2 . [3]

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(b) (i) Express S_{n+3} in terms of S_{n+2} and S_n . [1]

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(ii) Hence, or otherwise, find the value of S_4 . [2]

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- (c) Use the substitution $y = S_1 - x$, where S_1 is the numerical value found in part (a), to find and simplify an equation whose roots are $\alpha + \beta$, $\beta + \gamma$, $\gamma + \alpha$. [3]

[illegible]

- (d) Find the value of $\frac{1}{\alpha+\beta} + \frac{1}{\beta+\gamma} + \frac{1}{\gamma+\alpha}$. [2]

[illegible]